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ORB: A GOVERNANCE CONSOLE FOR LOCAL AND CLOUD LANGUAGE MODELS

By

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Specialization Project report submitted in partial fulfilment of the requirements of IV

Trimester MCA,

CHRIST (Deemed to be University)

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CERTIFICATE

*This is to certify that the report titled **Orb: A Governance Console for Local and Cloud Language Models** is a bona fide record of work done by Praneeth M (2547142), Parthiv Sushil (2547141) and Hari Prasad (2547120) of CHRIST (Deemed to be University), Bangalore, in partial fulfilment of the requirements of IV Trimester MCA during the year 2026*

Head of the Department

Project Guide

Valued by:

1. Name : Hari Prasad

2. Register Number : 2547120

Examination Centre : CHRIST (Deemed to be University)

Date of Exam :

AI USAGE DECLARATION STATEMENT

We hereby declare that Artificial Intelligence tools were used only for supportive purposes such as idea exploration, conceptual understanding, literature guidance, and grammar refinement during the development of this project. No AI generated code has been copied or submitted, and all programming, system development, experimentation, analysis, and results were independently developed and implemented by our team. We confirm that this project is our original work and reflects our own understanding and effort, and we take full responsibility for its authenticity and adherence to academic integrity policies.

We also declare that the approximate usage of Artificial Intelligence tools for supportive purposes in this project is 55%, limited to idea exploration, literature search guidance, grammar refinement, clarification of technical concepts, and other academic assistance.

The details of AI tools used in this project and their purpose are listed below:

1. **Name of the AI Tool:** Claude (Anthropic)
Purpose of Use: Ideation and conceptual assistance
2. **Name of the AI Tool:** ChatGPT (OpenAI)
Purpose of Use: Clarification of technical concepts
3. **Name of the AI Tool:** Grammarly
Purpose of Use: Grammar and spell check

This declaration confirms that the AI tools mentioned above were used only for supportive and assistive purposes and that no AI generated code has been directly copied or submitted as part of this project.

Name of the Student : Hari Prasad

Register Number : 2547120

Project Title : Orb: A Governance Console for Local and
Cloud Language Models

Signature :

Date :

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ABSTRACT

Large language models are increasingly executed on personal machines, where they can invoke tools that read files, run shell commands and access the web, yet users have little visibility into or control over these actions. Orb addresses this gap as a governance-first command console for local and cloud language models. The system places a governed Express/TypeScript gateway between the user and the model: it routes chat requests either to local Ollama models or to cloud providers (Anthropic, OpenAI, Groq, Mistral), enforces a per-tool execution policy of automatic, policy-based or manual approval, streams responses to a Next.js frontend, and persists sessions, audit logs, provider connector keys and extracted memory facts in SQLite. Every programmatic request through the Bearer-token API is fully audited with request messages, tool calls, policy decisions, latency and token counts. An asynchronous post-chat analysis stage extracts durable user facts into a per-user memory store and computes a hallucination-risk score for each response. Authentication is handled by Clerk for interactive users and by hashed API keys for external clients. The result is a working oversight layer that demonstrates how policy enforcement, auditability and telemetry can be added around both local and cloud model inference without modifying the models themselves.

This report documents the background, system analysis, requirement specification and design of Orb in its current state. It covers the existing-system landscape and its limitations, the proposed system and its features, the functional and non-functional requirements, the system architecture, data flow and database design, the interface design of the console, and the implementation, testing and results of the working system, followed by the conclusion, future enhancements, user manual and project planning.